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Since centripetal acceleration is $\frac{v^2}{r}$, the magnitude of this resultant acceleration is given by $a_R = \sqrt{a_t^2 + \left(\frac{v^2}{r}\right)^2}$.

This resultant acceleration makes an angle θ with the radius vector as shown in Figure 2.53.

This angle is given by $\tan\theta = \frac{a_t}{\left(v^2/r\right)}$.

EXAMPLE 2.40

A particle moves in a circle of radius 10 m. Its linear speed is given by $v = 3t$ where t is in second and v is in m s^{-1}.

(a) Find the centripetal and tangential acceleration at t = 2 s.

(b) Calculate the angle between the resultant acceleration and the radius vector.

Solution

The linear speed at t = 2 s

$$v = 3t = 6\,m\,s^{-1}$$

The centripetal acceleration at t = 2 s is

$$a_c = \frac{v^2}{r} = \frac{(6)^2}{10} = 3.6\,m\,s^{-2}$$

The tangential acceleration is $a_t = \frac{dv}{dt} = 3\,m\,s^{-2}$

The angle between the radius vector with resultant acceleration is given by

$$\tan\theta = \frac{a_t}{a_c} = \frac{3}{3.6} = 0.833$$

$$\theta = \tan^{-1}(0.833) = 0.69 \text{ radian}$$

In terms of degree $\theta = 0.69 \times 57.17° \approx 40°$

Are we moving with the same speed both day and night with respect to Sun?

Earth orbits the Sun in an elliptical orbit. Let us specify the velocity of the center of Earth with respect to Sun as $\vec{v}_c$. This $\vec{v}_c$ is due to the elliptical motion of the Earth around the Sun. We know that at the same time Earth is also spinning on its own axis. Due to this spinning, all objects on the surface of the Earth undergo circular motion with velocity $(\vec{v}_s)$ with respect to the axis of rotation of the Earth. At night both $\vec{v}_c$ and $\vec{v}_s$ are either in the same direction or at an acute angle with each other. So, the velocity of an object on the surface of Earth with respect to Sun at night time is $\vec{v}_{night} = \vec{v}_c + \vec{v}_s$. During the day $\vec{v}_c$ and $\vec{v}_s$ are either in opposite directions or at an obtuse angle with each other. So, the velocity of the object with respect to Sun at day time $\vec{v}_{day} = \vec{v}_c - \vec{v}_s$. From this, we can conclude that any object on the surface of the Earth travels faster with respect to Sun during night than during day time. This happens due to the rotation of the Earth.

